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CHB-101

Examination –May- 2024

B.Tech. I/II Sem : Common to EE,EC,IoT, CSE,CSEBC, IT, AIADS branch

Applied Chemistry

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

				BL	CO	PO	PI Code
Q.1	(a)	Choose the correct option. Only one answer is correct. Write the correct answer only.	02	1	1	2	2.5.1

(i) Chemical name of Calgon is

1. Sodium ortho phosphate
2. Sodium Potassium phosphate
3. Sodium hexa meta phosphate
4. Sodium hydrogen phosphate

(ii) Which ion is absent in naturally occurring zeolite?

1. Sodium ion 2. Calcium ion
3. Aluminium ion 4. Silicon ion

	(b)	In determination of hardness of water, why disodium salt of EDTA is taken in place of EDTA?	02	5	1	1	1.5.1
	(c)	Explain scale and sludge formation. Write disadvantages of scale and sludge formation.	03	1	1	1	1.5.1

OR

		100 ml of water sample consumed 20 ml of N/50 HCl for neutralization up to Phenolphthalein end point. At this point, methyl orange indicator was added and the same sample further required 20 ml of N/50 HCl. Find type and extent of alkalinity in water sample.	03	3	1	1	1.5.1
	(d)	Discuss Ion Exchange process of water softening giving diagram, chemical reaction and regeneration process.	05	2	1	1	1.5.1

OR

		Write short notes on: (i) Priming and Foaming (ii) Internal conditioning methods	05	2	1	1	1.5.1

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Q.2 (a) Choose the correct option. Only one answer is correct. Write the correct answer only. 02

1	2	2	2.5.1

(i) In a galvanic cell

1. Chemical reaction produces electrical energy
2. Electrical energy produces chemical reaction
3. Reduction occurs at anode
4. Oxidation occurs at cathode

(ii) Electrolyte used in Li Ion Batteries is

1. Ammonium chloride and Ammonium hydroxide
2. Gel polymer electrolyte
3. Zinc chloride
4. Manganese di oxide

(b) Write mathematical expression and importance of Nernst's equation. 02

5	2	1	1.5.1

(c) Discuss working of an electrochemical cell giving suitable example. 03

2	2	1	1.5.1

OR

Calculate the emf of cell reprented as 03

3	2	1	1.5.1

$Zn_{(s)} \mid ZnSO_{4(0.1M)} \parallel CuSO_{4(0.001M)} \mid Cu(s)$

Given $E^{\circ}_{Cell} = 1.103$ V at 25°C.

(d) Discuss construction and working of Lithium ion battery. Compare Li ion battery with Na ion battery. 05

6	2	1	1.5.1

OR

Write brief notes on: 05

2	2	1	1.5.1

(i) Single Electrode Potential

(ii) Secondary Batteries

Q.3 (a) Choose the correct option. Only one answer is correct. Write the correct answer only. 02

1	3	2	2.5.1

(i) Rusting is ..... process

1. Polymerisation
2. Redox
3. Condensation
4. Vaporisation

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(ii) Pilling Bedworth ratio is the ratio of

1. Volume of metal hydroxide to volume of metal
2. Volume of metal to volume of metal hydroxide
3. Volume of metal oxide to volume of metal
4. Volume of metal to volume of metal oxide

(b)	Iron corrodes faster than aluminium. Why?	02	5	3	1	1.5.1
(c)	What are alloys? Give their significance with suitable examples.	03	2	3	1	1.5.1

OR

	What is corrosion? What are disadvantages of corrosion?	03	2	3	1	1.5.1
(d)	Discuss various theories to illustrate the process of corrosion.	05	2	3	1	1.5.1

OR

	Write short notes on: (i) Electroplating (ii) Sacrificial protection	05	2	3	1	1.5.1
Q.4	(a) Choose the correct option. Only one answer is correct. Write the correct answer only. (i) Which of the following is not a polymer 1. Rubber 2. Protein 3. Sugar 4. Glass (ii) The measurement range of electron microscopy is around ..... 1. 10^{-9} m 2. 10^{9} cm 3. 10^{9} m 4. 10^{-9} cm	02	1	4	2	2.5.1
(b)	What are optical fibers? Write its uses.	02	5	4	1	1.5.1
(c)	What are nano particles? How are they synthesised?	03	4	4	1	1.5.1

OR

	Write an informatory note of Electroactive Polymers.	03	2	4	1	1.5.1

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(d) What are photovoltaic materials? Write principle and applications of dye sensitised solar cells. 05

2	4	1	1.5.1

OR

Write short notes on following: 05

1. Liquid Crystal Polymers
2. Graphene

2	4	1	1.5.1

Q.5 (a) Choose the correct option. Only one answer is correct. Write the correct answer only. 02

(i) Which wavelength range corresponds to uv-vis range of electro magnetic spectrum

1. 1 to 200 nm
2. 200 to 400 nm
3. 400 to 800 nm
4. 200 to 800 nm

(ii) Which of the following is a separation technique

1. Conductivitymetry
2. Colorimetry
3. IR spectroscopy
4. Chromatography

1	5	2	2.5.1

(b) Define pH. Calculate pH value of 0.01 M solution of Sodium Hydroxide 02

5	5	1	1.5.1

(c) State and explain Beer Lambert's law of colorimetry. 03

2	5	1	1.5.1

OR

What is chromatography? Give classification of chromatography with suitable examples. 03

2	5	1	1.5.1

(d) Discuss principle, instrumentation and applications of conductivitymetry. 05

2	5	1	1.5.1

OR

Write short notes on: 05

(i) IR spectroscopy

(ii) Importance of instrumental techniques.

2	5	1	1.5.1

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